

# Manger agent features

Operational Intelligence & Multi-Agent Orchestration Platform

## Core Features

*Build these — this forms the load-bearing foundation of the entire architecture.*

### 1. Multi-Agent Recommendation Aggregation

This is the Manager's foundational job: it receives four independent recommendations (one from each department agent) for the same incident and merges them into a single coherent plan. Mechanically, each agent's output should be structured the same way (recommended action, reasoning, confidence, constraints) so the Manager can compare them like-for-like rather than parsing four different formats. Without this, you don't have "one AI system," you have four separate chatbots that happen to run in parallel — this feature is what actually makes it one system.

### 2. Conflict Detection & Negotiated Resolution

Before merging, the Manager checks whether the four recommendations actually agree. When they don't (Front Desk wants to comp a VIP a free upgrade, Revenue flags that the same room was about to sell at a premium), the Manager doesn't just pick one side — it looks for a compromise that satisfies the underlying goal of both agents (e.g., comp a lesser amenity instead of the room, or backfill the premium room with another guest). This is the single most convincing thing you can show a judge, because it's the difference between "the AI aggregates opinions" and "the AI actually reasons under disagreement."

### 3. Confidence Scoring (Per Action + Composite)

Each department agent attaches its own certainty to its recommendation (e.g., Maintenance is 95% sure the AC needs a technician; Revenue is 60% sure moving the VIP won't hurt tonight's yield). The Manager combines these into a composite confidence score for the overall plan — not a simple average, but weighted by how critical each agent's domain is to this specific incident (Maintenance's certainty matters more when

the incident is equipment failure; Revenue's matters more when it's a pricing decision). This composite score is what later features (escalation, explainability) both depend on.

#### 4. Explainable Plan Generation

For every plan the Manager produces, it generates a structured breakdown answering:

- **Actionable Steps:** What concrete tasks need executing
- **Reasoning Chain:** Why this plan was chosen (including inputs from contributing agents)
- **Stakeholders:** Which departments are involved
- **Expected Impact:** Estimated metrics on cost, guest satisfaction, and operational resolution time

This isn't just a nice-to-have — it's what turns your demo from "the AI made a decision" into "here's exactly how and why the AI made this decision," which is the strongest form of trust-building you can show judges evaluating an AI-in-operations product.

#### 5. Human-in-the-Loop Escalation Gate

This decides whether a plan auto-executes or waits for a human. The threshold should combine two variables, not one: confidence (how sure the agents are) and impact/blast-radius (how costly or hard-to-reverse the action is, and who it affects — a VIP guest raises the bar). A high-confidence, low-impact action (reassigning a standard room) can auto-execute; a high-confidence, high-impact action (comping a VIP's entire stay) still needs a human's eyes, because confidence isn't the same as low risk. This is the feature that makes your "human stays in control" claim actually mean something, rather than just being a slide bullet.

#### 6. Task Dispatch & Real-Time Execution Handoff

Once a plan is approved (by a human, or auto-approved under the gate above), the Manager breaks it into discrete, department-routed tasks, assigns them, and pushes state updates live to the dashboard (via your WebSocket layer). This is the "close the loop" feature — it's what proves the system doesn't stop at recommending, it actually does something, which was the explicit requirement in your problem statement ("go beyond simply displaying data").

## 7. Aggregate Sentiment & Systemic Issue Pattern Detection

Rather than only reacting to single incidents, the Manager also rolls up signals across many guests over time — e.g., if the happiness score dips for several unrelated guests around the same time window, or the same complaint keyword (slow check-in, noisy AC) recurs across rooms, it surfaces this as a systemic alert rather than treating each as an isolated case ("14 guests this week flagged slow check-in — this looks like a staffing issue, not an individual guest issue"). This is a cheap addition since it reuses data the Front Desk and Housekeeping agents already produce, and it directly plugs the property-wide sentiment-analysis capability your problem statement asks for, which none of your other four agents currently own.

### Optional Features

*Build if time allows — these add depth and serve as strong "if you ask, I have more" answers during Q&A.*

| Feature Name                                                  | Functionality & Scope                                                                                                                              | Operational Value                                                                                                      |
|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Historical Decision Memory &amp; Outcome Feedback Loop</b> | Records past decisions and outcomes (e.g., guest happiness recovery, execution timeline) to dynamically adjust future agent confidence weightings. | Demonstrates a self-improving system narrative using lightweight weighted-history logic rather than complex RL models. |
| <b>Escalation Tiering / Stakeholder Routing</b>               | Routes pending approvals to specific organizational roles based on severity (e.g., Shift Supervisor vs. General Manager).                          | Replaces a flat approval gate with realistic enterprise administrative hierarchy.                                      |
| <b>Natural-Language Shift Briefing</b>                        | Generates narrative summaries of overnight events, resolved incidents,                                                                             | Enhances shift handoffs and highlights practical quality-of-life workflow                                              |

| Feature Name                                 | Functionality & Scope                                                                                             | Operational Value                                                                                 |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|                                              | and pending actions at shift change.                                                                              | improvements for human staff.                                                                     |
| <b>Decision Audit Trail / Compliance Log</b> | Maintains an immutable log of proposed plans, modifications, timestamps, and human vs. automated approval events. | Directly addresses enterprise accountability, traceability, and operational compliance inquiries. |